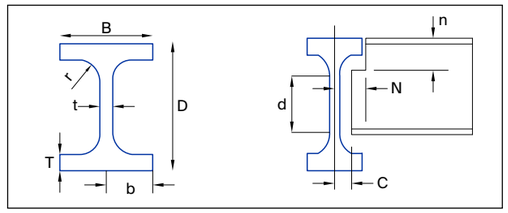
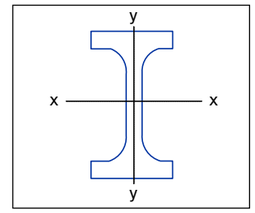
2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
350x175	41.2	346	174	9	6	13	302	52.5	9.67	50.3	5	94	22
	41.4	346	174	9	6	14	300	52.7	9.67	50.0	5	94	23
	49.4	350	175	11	7	13	302	62.9	7.95	43.1	6	94	24
	49.6	350	175	11	7	14	300	63.1	7.95	42.9	6	94	25
	57.8	354	176	13	8	14	300	73.7	6.77	37.5	6	94	27
	66.2	358	177	15	9	14	300	84.3	5.90	33.3	7	94	29
	71.8	360	178	16	10	14	300	91.4	5.56	30.0	7	94	30
	79.7	364	177	18	11	14	300	101	4.92	27.3	8	93	32
350x250	69.2	336	249	12	8	20	272	88.2	10.4	34.0	6	131	32
	78.1	340	250	14	9	13	286	100	8.93	31.8	7	131	27
	79.7	340	250	14	9	20	272	102	8.93	30.2	7	131	34
	94.2	346	251	17	10	20	272	120	7.38	27.2	7	131	37
	108	350	253	19	12	20	272	137	6.66	22.7	8	131	39
350x350	113	344	348	16	10	13	286	144	10.9	28.6	7	179	29
	115	344	354	16	10	20	272	148	11.1	27.2	7	182	36
	135	350	350	19	12	13	286	172	9.21	23.8	8	179	32
	137	350	350	19	12	20	272	174	9.21	22.7	8	179	39
	159	356	352	22	14	20	272	202	8.00	19.4	9	179	42
	181	362	354	25	16	20	272	230	7.08	17.0	10	179	45
400x200	56.1	396	199	11	7	13	348	71.4	9.05	49.7	6	106	24
	56.6	396	199	11	7	16	342	72.2	9.05	48.9	6	106	27
	65.4	400	200	13	8	13	348	83.4	7.69	43.5	6	106	26
	66.0	400	200	13	8	16	342	84.1	7.69	42.8	6	106	29
	75.5	404	201	15	9	16	342	96.2	6.70	38.0	7	106	31
	88.2	410	202	18	10	16	342	112	5.61	34.2	7	106	34
	140	430	208	28	16	16	342	179	3.71	21.4	10	106	44
	187	446	214	36	22	16	342	239	2.97	15.5	13	106	52
400x300	94.3	386	299	14	9	22	314	120	10.7	34.9	7	155	36
	105	390	300	16	10	13	332	133	9.38	33.2	7	155	29
	107	390	300	16	10	22	314	136	9.38	31.4	7	155	38
	127	396	302	19	12	22	314	162	7.95	26.2	8	155	41
	144	402	303	22	13	22	314	184	6.89	24.2	9	155	44

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.35	11040	792	14.5	3.88	638	91.0	712	140	0.881	38.1	0.225	13.2
1.35	11100	792	14.5	3.88	641	91.0	716	140	0.882	37.5	0.225	13.6
1.36	13500	984	14.6	3.96	771	113	864	173	0.883	32.2	0.283	22.4
1.36	13560	985	14.7	3.95	775	113	868	174	0.884	31.8	0.283	23.0
1.37	16100	1184	14.8	4.01	909	135	1022	208	0.885	27.6	0.344	36.1
1.38	18710	1390	14.9	4.06	1045	157	1180	243	0.886	24.4	0.409	53.6
1.39	20240	1508	14.9	4.06	1124	169	1276	263	0.883	22.9	0.446	66.3
1.39	22760	1669	15.0	4.06	1250	189	1425	293	0.884	20.6	0.499	91.3
1.62	18510	3093	14.5	5.92	1102	248	1215	380	0.881	25.8	0.812	44.6
1.64	21230	3649	14.6	6.05	1249	292	1382	445	0.881	24.2	0.969	58.0
1.63	21680	3652	14.6	6.00	1275	292	1412	447	0.882	22.8	0.970	66.3
1.64	26440	4488	14.8	6.12	1528	358	1699	547	0.886	19.5	1.21	109
1.65	30190	5138	14.8	6.12	1725	406	1935	623	0.882	17.6	1.41	155
2.04	32850	11240	15.1	8.84	1910	646	2092	978	0.848	21.1	3.02	111
2.05	33810	11840	15.1	8.95	1966	669	2153	1014	0.846	20.4	3.18	123
2.05	39847	13583	15.2	8.89	2277	776	2515	1176	0.847	18.0	3.72	186
2.04	40300	13590	15.2	8.84	2303	776	2545	1179	0.849	17.5	3.72	199
2.06	47590	16000	15.3	8.90	2674	909	2979	1382	0.848	15.4	4.46	305
2.07	55190	18500	15.5	8.96	3049	1045	3424	1591	0.848	13.7	5.25	445
1.55	19770	1447	16.6	4.50	999	145	1114	223	0.883	36.8	0.536	25.0
1.55	20020	1448	16.7	4.48	1011	145	1128	224	0.884	35.6	0.536	27.1
1.56	23460	1736	16.8	4.56	1173	174	1313	267	0.885	31.8	0.650	39.6
1.56	23710	1737	16.8	4.54	1185	174	1326	268	0.886	30.9	0.650	42.2
1.57	27490	2035	16.9	4.60	1361	202	1528	312	0.887	27.4	0.770	62.3
1.58	33060	2478	17.2	4.70	1612	245	1815	378	0.891	23.4	0.952	101
1.63	54850	4216	17.5	4.86	2551	405	2941	632	0.887	15.7	1.70	373
1.68	75250	5919	17.8	4.98	3374	553	3968	873	0.883	12.5	2.49	825
1.91	33680	6246	16.7	7.21	1745	418	1918	637	0.880	25.8	2.16	79.9
1.94	37860	7204	16.9	7.35	1942	480	2141	730	0.879	24.5	2.52	100
1.92	38680	7210	16.9	7.28	1983	481	2188	733	0.881	23.2	2.52	114
1.94	46660	8735	17.0	7.35	2357	578	2620	884	0.880	20.0	3.10	185
1.95	54420	10220	17.2	7.45	2707	674	3022	1030	0.883	17.7	3.69	272